

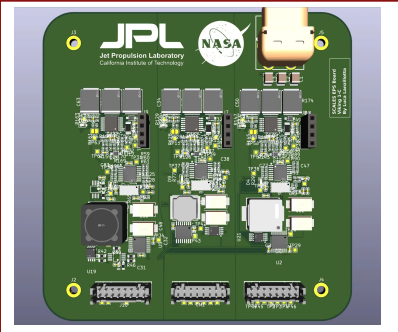
Board Description

This merger board seeks to combine the SCALES EPS board with the SCALES IMX8 Carrier Board. Both boards have undergone three revisions, this merger will combine the final revisions of both with each of them undergoing some more modifications based on separate board reviews. Notable differences for each board compared to their individual counterparts are listed below.

EPS: Based on the Rev F Schematic / Viking 1-C model, (Modifications based on suggestions by A. Greenberg)
Watchdog removed for Peripheral Board
More decoupling capacitors for inrush current mitigation for Jetson bootup sequence
Resistors to bias EN lines for the perif/jetson
Resistors to current limit Watchdog outputs

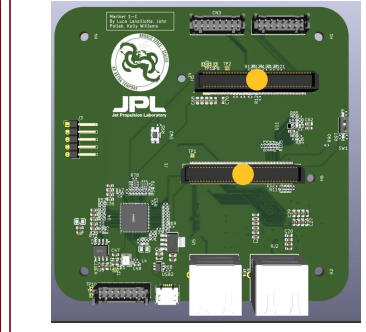
IMX8 Carrier Board: Based on the Mariner 1-C board (Modifications suggested by Phytex)
- Drop LSIO_QSPI0B-DATA 0 to 3, exclude from Pinmux (Pg. 2)
- Consider Pgood implementation for i2c0 and i2c3, spi2, and uart
- For USB2, remove leave R68 Not mounted (Pg.5) (verified on pcm 942)
- Eth1 RX pull up R18, R20 and R21 leave Not Mounted (Pg. 6)
- VDDA1P8 Change VCC_ETH_1V0 to VCC_1V8
- For Pgood, Isolate SOM 3v3, and use an load switch for the 3v3 peripheral systems and a switching regulator with an enable line triggered by pgood for the 1.8v system
- Using TI TLV62569PDDCR for switching and sequencing from 3.3v to 1.8v using Pgood
- Using TPS22917 for 3.3v peripheral power sequencing using Pgood
- Remove 74AUP1G08 and tie POR_B_1V8 through a 15k resistor to Reset_N
- Reset button pull up to SOM VCC_3V3 not VCC3V3OUT

SCALES_EPS



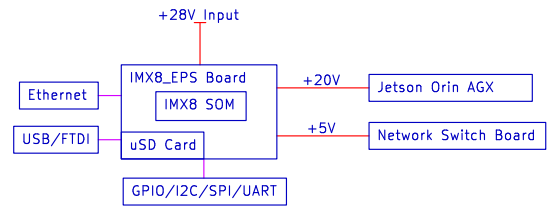
File: SCALES_EPS.kicad_sch

IMX8QXP SCALES Mariner Carrier Board



File: imx8qxp_carrier_board.kicad_sch

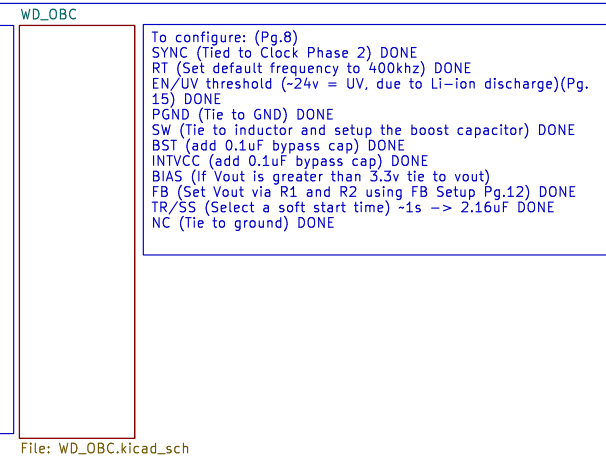
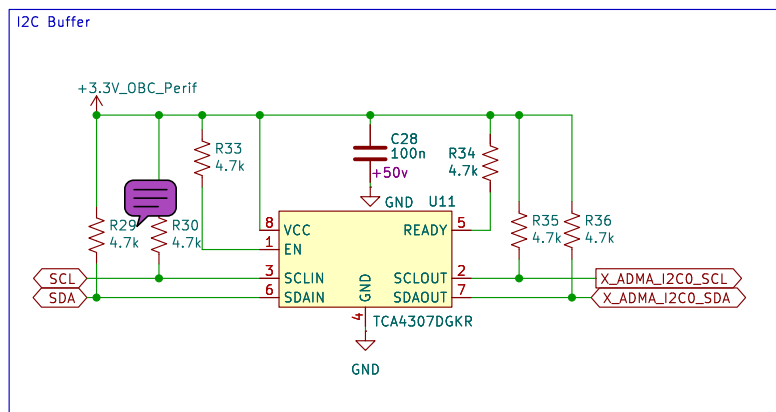
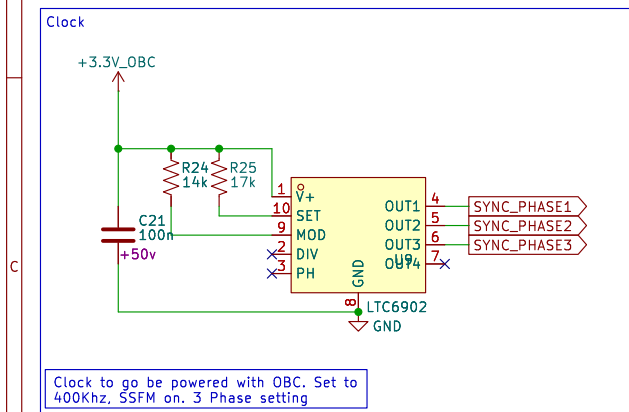
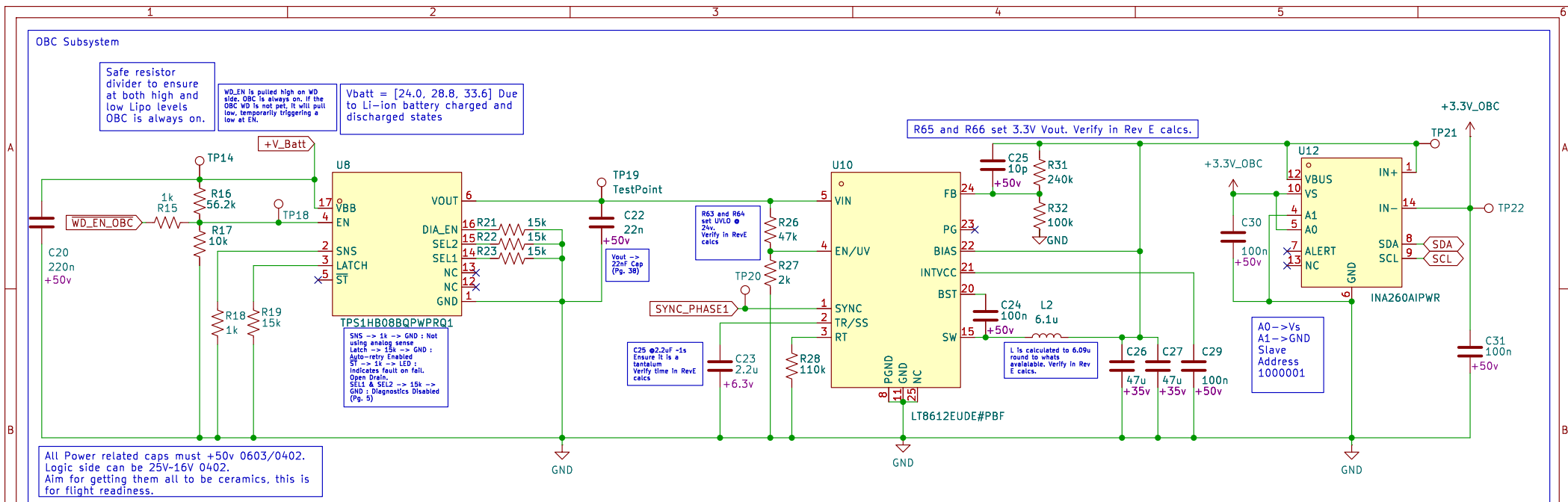
Block Diagram



By Luca Lanzillotta & John Pollack
Bronco Space Lab
Sheet: /
File: imx8x_eps_leviathan.kicad_sch

Title: Leviathan 1-A

Size: A5	Date: 2025-10-29	Rev: 1-A
KiCad E.D.A. 9.0.3		Id: 1/16



By Luca Lanzillotta & John Pollack

Bronco Space Lab

Sheet: /SCALES_EPS/OBC_Subsystem/

File: OBC_Subsystem.kicad_sch

Title: Leviathan 1-A

Size: A4 Date: 2025-10-29

KiCad E.D.A. 9.0.3

Rev: 1-A

Id: 5/16

External Watch-Dog Timer

300uF Capacitance used to keep WD output high after reset for 1s to allow for subsystem power cycle

Disable shutdown during initial power up.

Pulled down. Holds Jetson off until toggled by OBC_JETSON_EN is held high by the OBC

Select 15V-25V on C21, C22, C23

~ 24 s period (36 s on power up)

C112+C114 and R62+R63 set the WDT frequency

Not using the recommended diode due to lack of stock. The DIP is almost exactly the same.

TLV1704AMPWPSEP is the radiation tolerant version. Use TLV1704AQPWRQ1 for non-flight C3 boards.

WD_EN_Perif goes to EN of the load switch controller

By Luca Lanzillotta & John Pollack

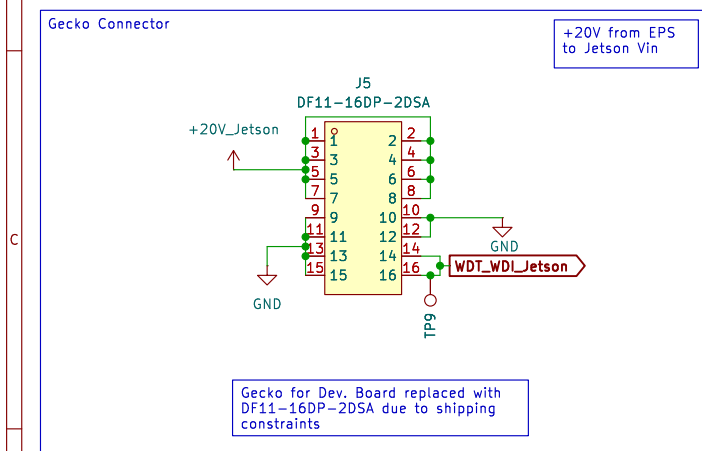
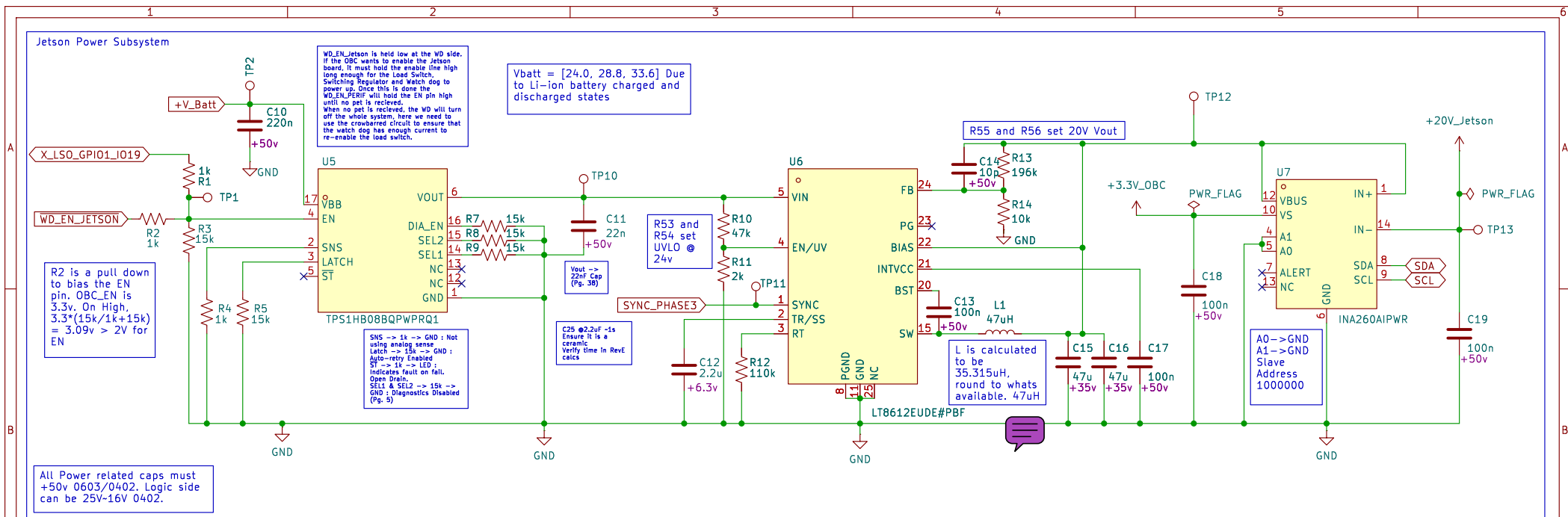
Bronco Space Lab

Sheet: /SCALES_EPS/OBC_Subsystem/WD_OBC/
File: WD_OBC.kicad_sch

Title: Leviathan 1-A

Size: A4 Date: 2025-10-29
KiCad E.D.A. 9.0.3

Rev: 1-A
Id: 7/16



To configure: (Pg.8)
 SYNC (Tied to Clock Phase 2) DONE
 RT (Set default frequency to 400khz) DONE
 EN/UV threshold (~24v = UV, due to Li-ion discharge)(Pg. 15) DONE
 PGND (Tie to GND) DONE
 SW (Tie to inductor and setup the boost capacitor) DONE
 BST (add 0.1uF bypass cap) DONE
 INTVCC (add 0.1uF bypass cap) DONE
 BIAS (if Vout is greater than 3.3v tie to vout)
 FB (Set Vout via R1 and R2 using FB Setup Pg.12) DONE
 TR/SS (Select a soft start time) ~1s $\rightarrow$ 2.16uF DONE
 NC (Tie to ground) DONE



By Luca Lanzillotta & John Pollack

Bronco Space Lab

Sheet: /SCALES_EPS/JetsonSubsystem/

File: JetsonSubsystem.kicad_sch

Title: Leviathan 1-A

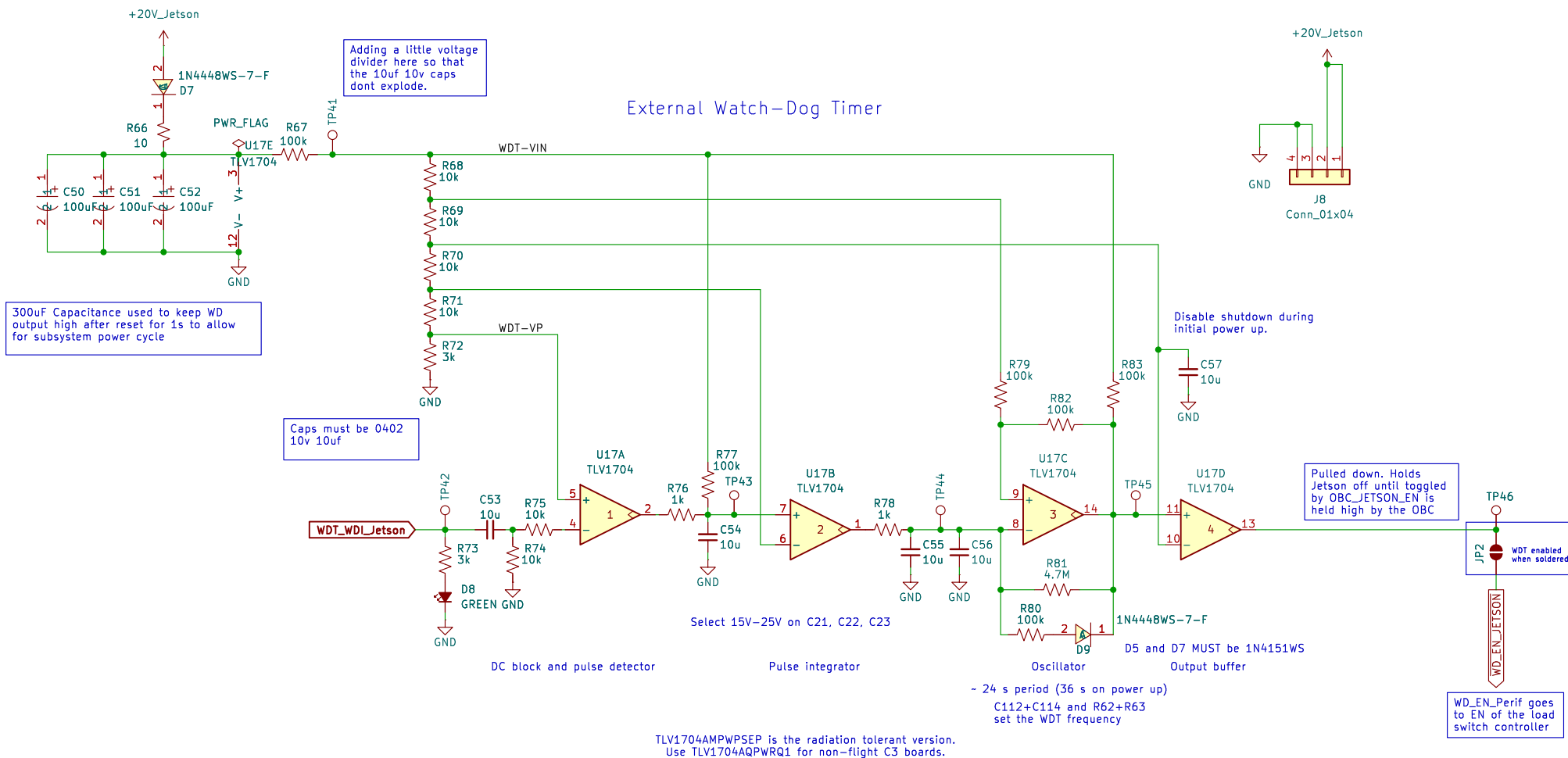
Size: A4 Date: 2025-10-29

KiCad E.D.A. 9.0.3

Rev: 1-A

Id: 4/16

External Watch-Dog Timer



By Luca Lanzillotta & John Pollack

Bronco Space Lab

Sheet: /SCALES_EPS/Jetson_Subsystem/WD_Jetson/

File: WD_Jetson.kicad_sch

Title: Leviathan 1-A

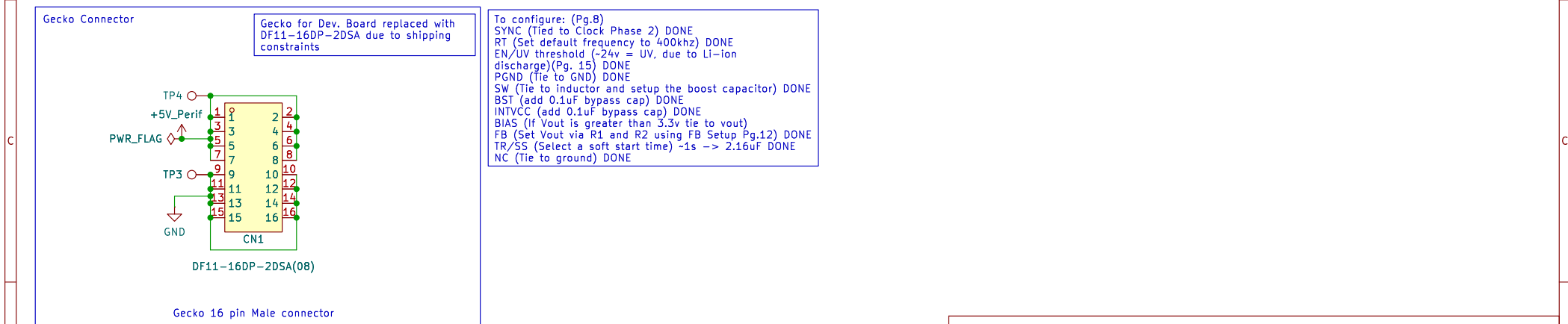
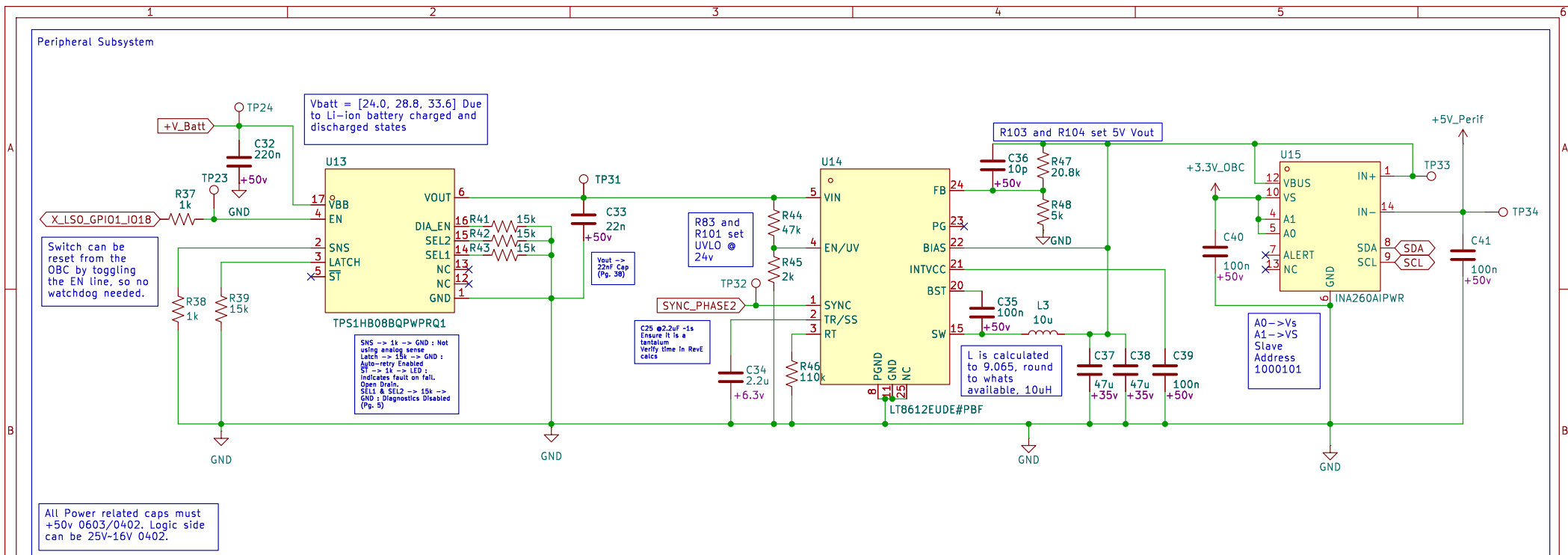
Size: A4

Date: 2025-10-29

KiCad E.D.A. 9.0.3

Rev: 1-A

Id: 8/16



By Luca Lanzillotta & John Pollack

Bronco Space Lab

Sheet: /SCALES_EPS/Perif_Subsystem/

File: Perif_Subsystem.kicad_sch

Title: Leviathan 1-A

Size: A4

Date: 2025-10-29

KiCad E.D.A. 9.0.3

Rev: 1-A

Id: 6/16

IMX8 SCALES Merger Board

Variant: Pre-Flight Testing Board

The IMX8 Scales Merger Board is a minimized version of the IMX8 Dev board combined with the SCALES EPS Board

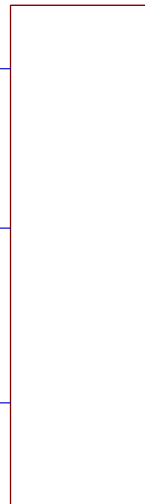
Board Description:

Provided serial communications for end user: FTDI, UART, SPI, I2C
Power Input: From SCALES EPS Rev F (Hirose DF11 2x8 Connector)
Ethernet: 2, 10/100 Base-T RJ-45 Connectors
SPI/GPIO/I2C: Provided via Hirose DF11 2x8 Connector
SOM Connector: 2x140 proprietary Samtec Connector for IMX8QXP SOM

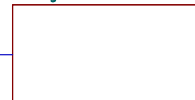
Ethernet



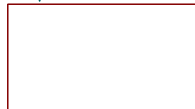
SOM



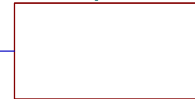
Debug USB - UART



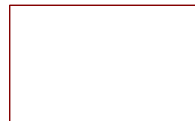
Peripheral Connections



Boot Configuration



Power



File: Pgood_Power.kicad_sch

By Luca Lanzillotta & John Pollack

Bronco Space Lab

Sheet: /IMX8QXP SCALES Mariner Carrier Board/

File: imx8qxp_carrier_board.kicad_sch

Title: Leviathan 1-A

Size: A5

Date: 2025-10-29

Rev: **1-A**

KiCad E.D.A. 9.0.3

Id: 3/16

10/100/1000 Ethernet

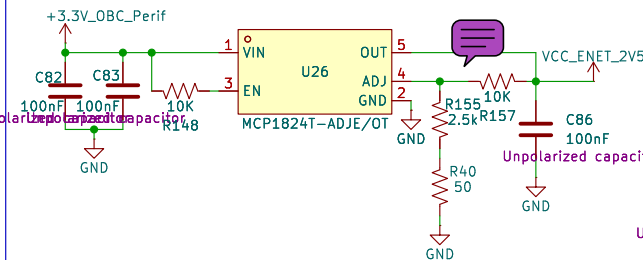
Description: Ethernet configuration and bootstrap modes are configured on the SOM.
This is all ported from the IMX8X dev board. There are two available Ethernet configurations for the SOM, one is preconfigured on the SOM with an onboard RGMII translator, and the second one is included in this carrier board.

Anything going to the SOM is single ended. 50ohm. Impedance Controlled.

3v3 dependent on Pgood. From Load switch

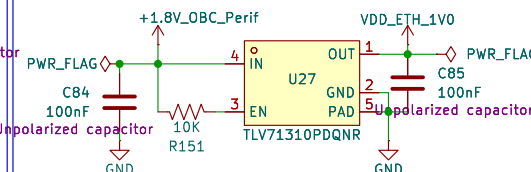
3v3 dependent on Pgood. From Load switch

VDD_ETH_2V5_SUPPLY

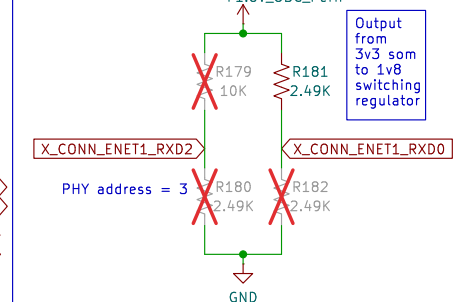


Output from 3v3 som to 1v8 switching regulator

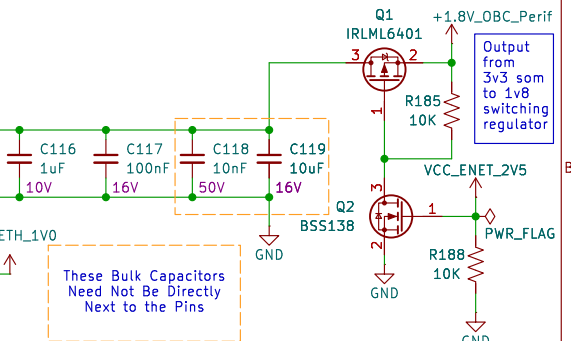
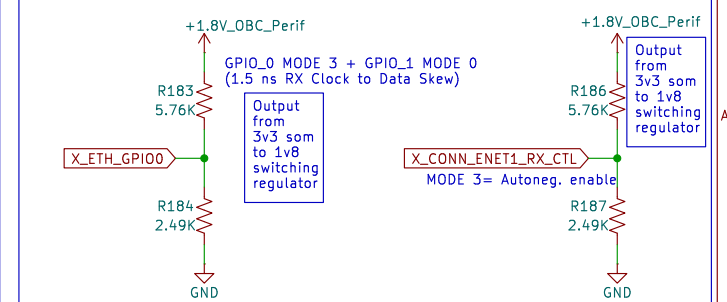
VDD_ETH_1V0_SUPPLY



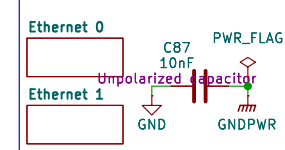
ETH1 RX Pull-up



ETH1 GPIO/RX Pull-up



Ethernet Ports



By Luca Lanzillotta & John Pollack

Bronco Space Lab

Sheet: /IMX8XQXP SCALES Mariner Carrier Board/Ethernet/

File: ethernet.kicad_sch

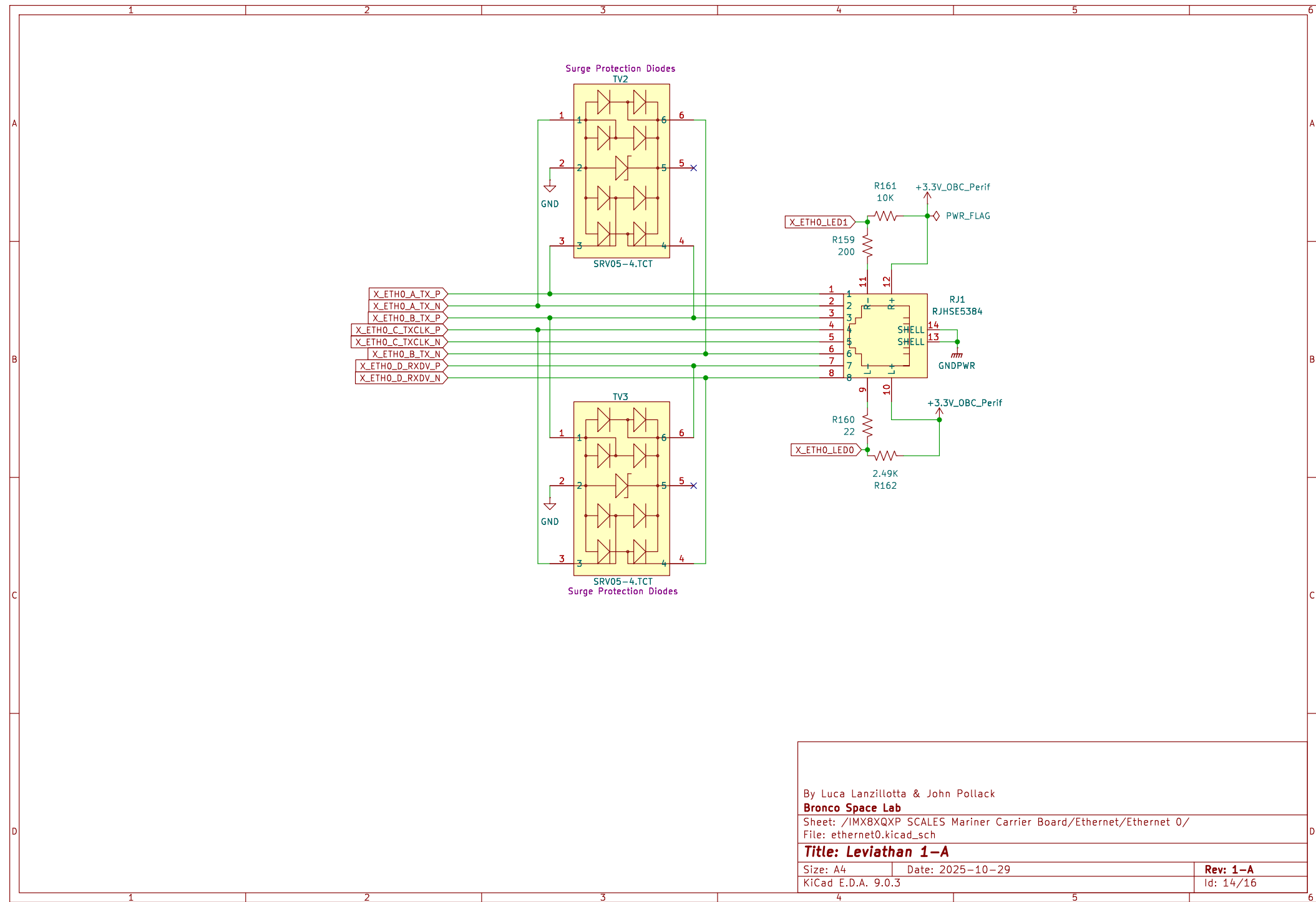
Title: Leviathan 1-A

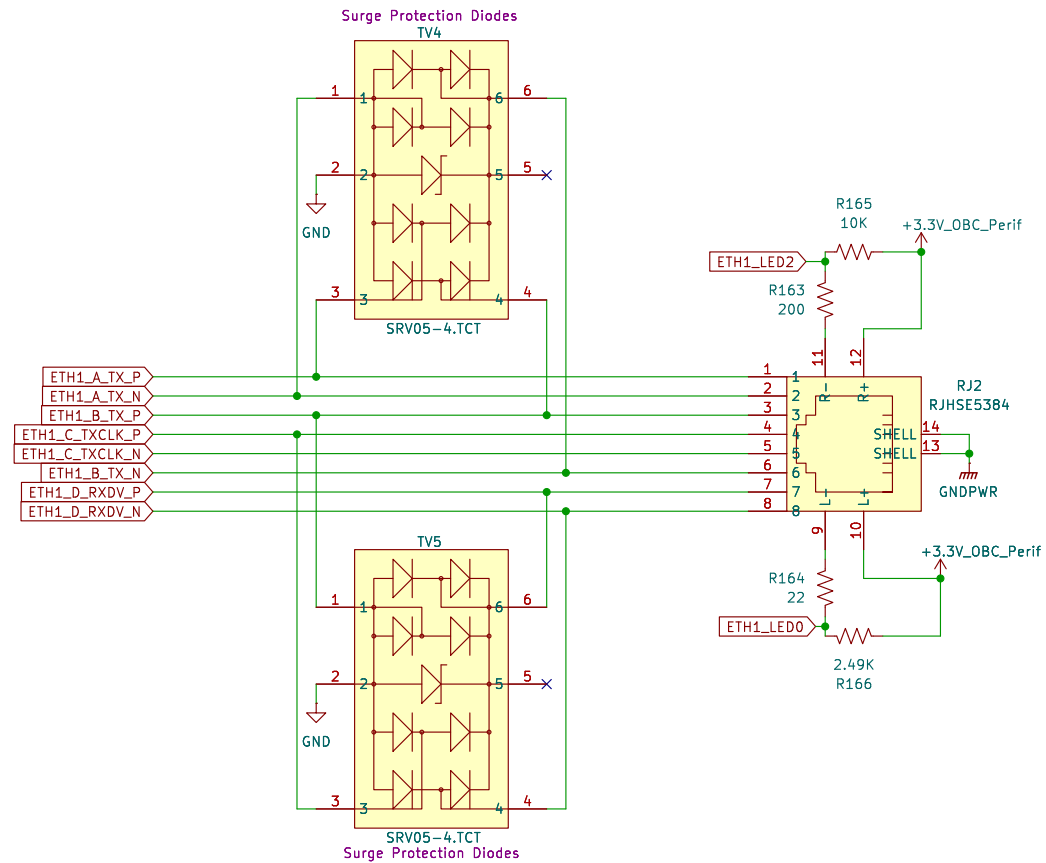
Size: A4 Date: 2025-10-29

KiCad E.D.A. 9.0.3

Rev: 1-A

Id: 12/16





By Luca Lanzillotta & John Pollack

Bronco Space Lab

Sheet: /IMX8XQXP SCALES Mariner Carrier Board/Ethernet/Ethernet 1/
File: ethernet1.kicad_sch

Title: Leviathan 1-A

Size: A4

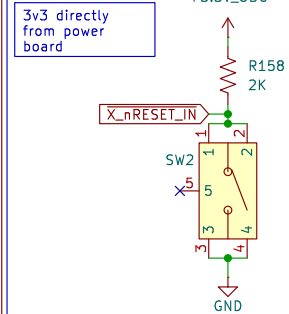
Date: 2025-10-29

Rev: 1-A

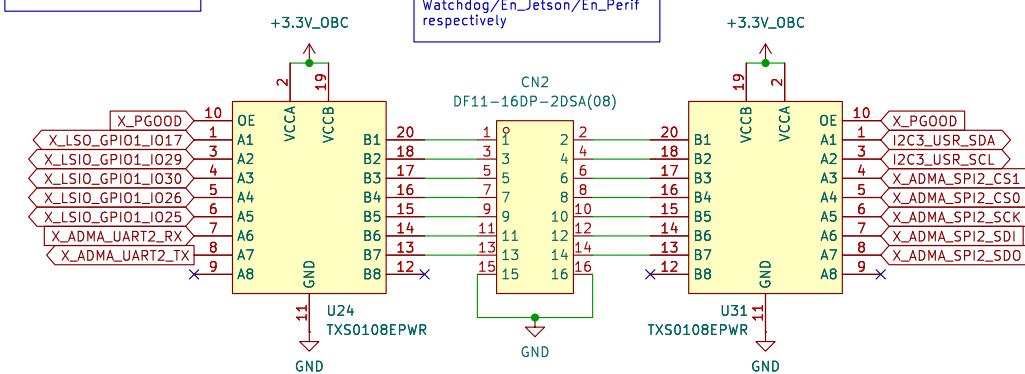
KiCad E.D.A. 9.0.3

Id: 15/16

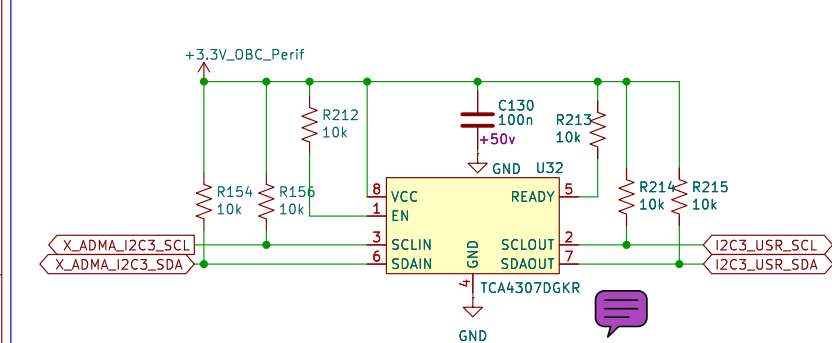
RESET BUTTON



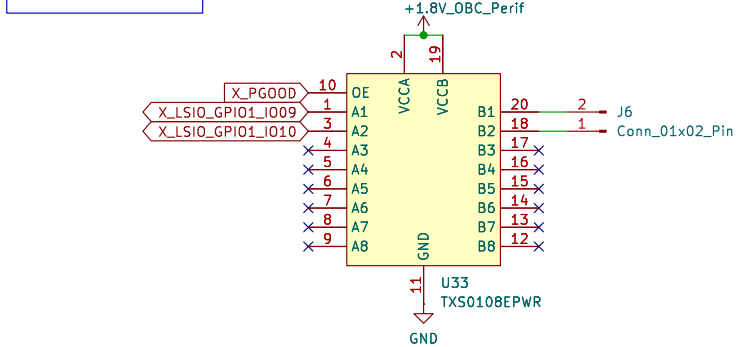
GPIO 3.3V PGood Buffer



I2C Buffer



GPIO 1.8V PGood Buffer



By Luca Lanzillotta & John Pollack

Bronco Space Lab

Sheet: /IMX8XQXP SCALES Mariner Carrier Board/Peripheral Connections/

File: periphCon.kicad_sch

Title: Leviathan 1-A

Size: A4

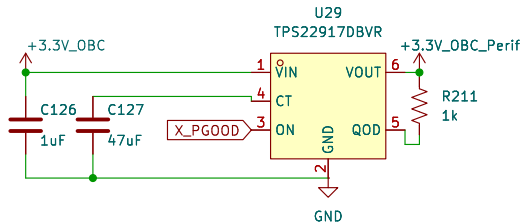
Date: 2025-10-29

Rev: 1-A

KiCad E.D.A. 9.0.3

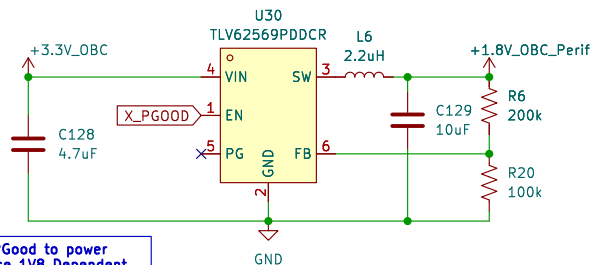
Id: 13/16

3V3_OBC to VCC_3V3 Load Switch



Using PGood to power sequence 3V3 Dependent peripheral devices on the carrier board.

3V3_OBC to VCC_1V8 Switching Regulator



Using PGood to power sequence 1V8 Dependent peripheral devices on the carrier board.

By Luca Lanzillotta & John Pollack

Bronco Space Lab

Sheet: /IMX8XQXP SCALES Mariner Carrier Board/Power/

File: Pgood_Power.kicad_sch

Title: Leviathan 1-A

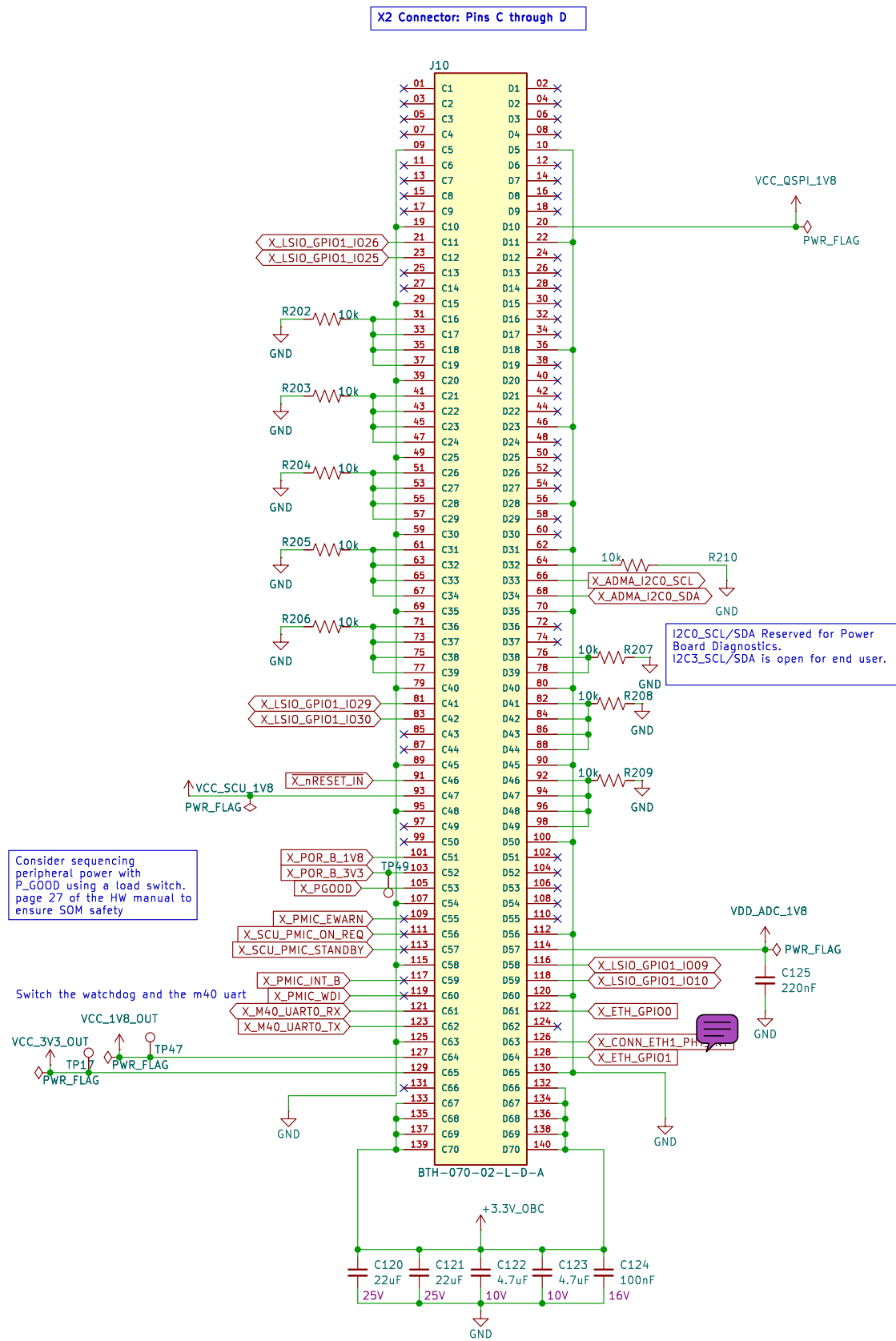
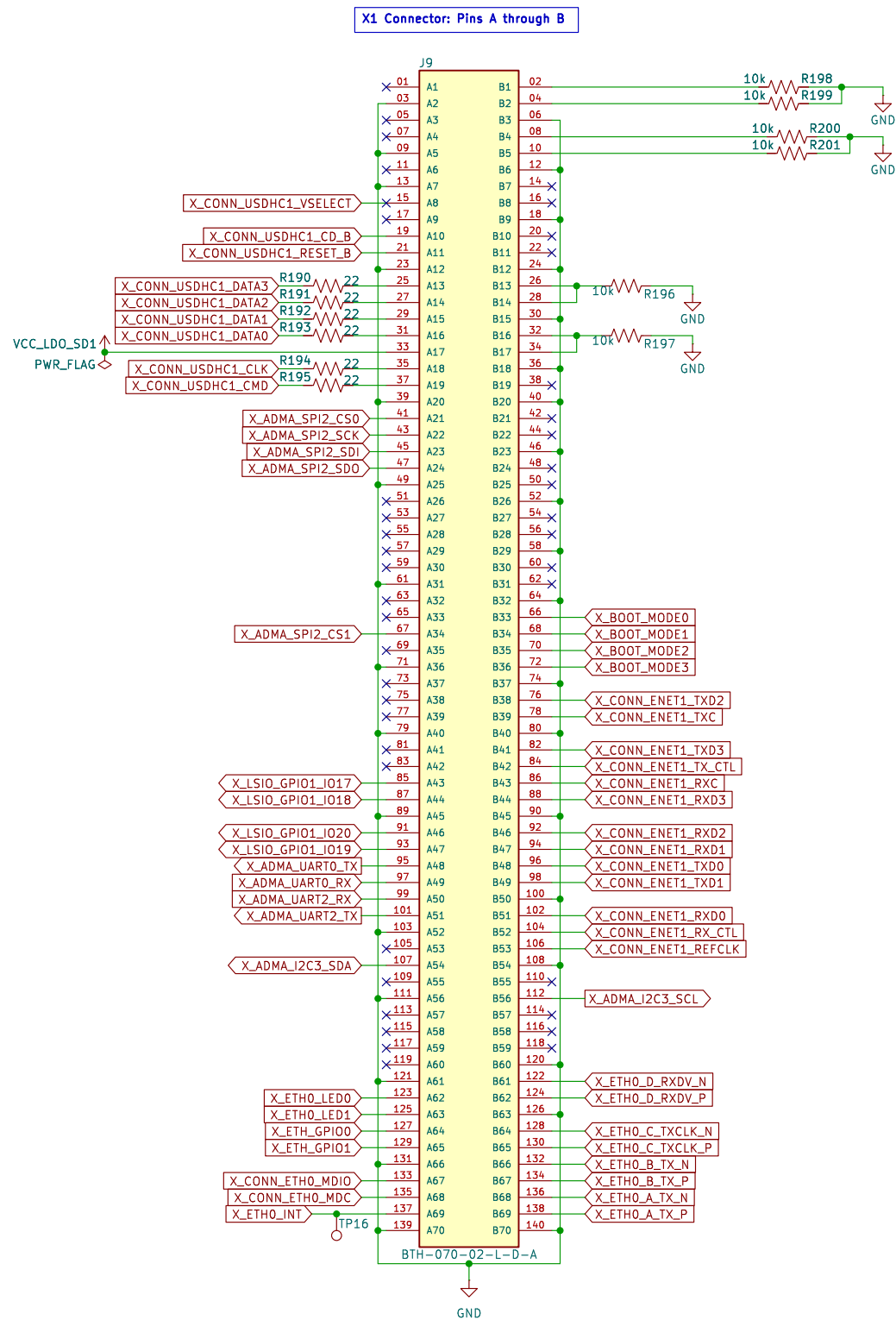
Size: A5

Date: 2025-10-29

Rev: 1-A

KiCad E.D.A. 9.0.3

Id: 16/16



The SDCard/Wifi Mux(s) have been removed to allow for Default SDCard boot configuration and Utilization. If it is desired to use eMMC, then there must be a switch to toggle between boot modes, this is BOOTMODE[3:0] bits 0010 to eMMC and 0011 to SDCard respectively. Due to removal of all other boot configurations, we only need to switch the last bit using the provided switch we will include.

The diagram illustrates the SD1 Power Enable Switch circuit. A +3.3V_OBC supply is connected to a 100nF capacitor. The capacitor is connected to the A1 pin of the BD2204GUL-E2 MOSFET. The gate of the MOSFET is also connected to the capacitor and the X_CONN_USDHC1_RESET_B signal. The MOSFET's source is connected to GND, and its drain is connected to the B1 pin, which is labeled SD1_SPWR. The MOSFET's VIN1 and VIN2 pins are connected to the gate, and its VOUT pin is connected to the drain. The MOSFET's GND and SEL pins are connected to GND. The MOSFET's B2 and B3 pins are also connected to GND. A callout box indicates that the MOSFET is a 'Unidirectional capacitor'.

The schematic diagram illustrates the electrical connections for a USB Card Connector. It shows the connection of a USB Type-A plug (pins 1-14) to a Micro-USB B connector (pins 1-14). The diagram includes the following components and connections:

- Power and Ground:**
 - VCC_LDO_SD1** is connected to pin 1 (DAT2).
 - +3.3V_D** is connected to pin 2 (DAT1).
 - VCC_SCU_1V8** is connected to pin 3 (CMD).
 - GND** is connected to pins 4 (VDD), 5 (CLK), 6 (VSS), 10 (SW), 11 (DET), 12 (SHELL), 13 (SHELL), and 14 (SHELL).
- Data and Control:**
 - X_CONN_USDHC1_DATA2** is connected to pin 1 (DAT2).
 - X_CONN_USDHC1_DATA3** is connected to pin 2 (DAT1).
 - X_CONN_USDHC1_CMD** is connected to pin 3 (CMD).
 - X_CONN_USDHC1_CLK** is connected to pin 5 (CLK).
 - X_CONN_USDHC1_DATA0** is connected to pin 7 (DAT0).
 - X_CONN_USDHC1_DATA1** is connected to pin 8 (DAT1).
 - X_CONN_USDHC1_CD_B** is connected to pin 9 (SW).
- Resistors:**
 - R89, R90, R91, R92, R93, R94, R95, R96, R97, R98, R99** are 49.9K resistors connected to the data and control lines.
 - R100** is a 0 ohm resistor connected to the VCC_SCU_1V8 line.
 - R101** is a 0 ohm resistor connected to the VCC_SCU_1V8 line.
- Capacitors:**
 - C59** and **C60** are 10uF unipolar capacitors connected to the VCC_SCU_1V8 line.
- Callout:** A callout box highlights the **+3.3V_D** connection to pin 2 (DAT1).

Id: 10/16